Science - K-12 Level Assessment 22 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. To produce light, the atoms inside a light bulb transform electrical energy into what other kind of energy?
   - (a) heat
   - (b) chemical
   - (c) mechanical
   - (d) electromagnetic
2. Which of the following is the primary type of stress that a fixed vise exerts on a piece of wood?
   - (a) tensile
   - (b) shear
   - (c) torsion
   - (d) compression
3. Why is water from an aquifer more likely to be cleaner than water from other sources?
   - (a) because it forms where fresh and salt water meet
   - (b) because it receives water directly from precipitation
   - (c) because it rises to the surface near the ocean
   - (d) because pollutants are filtered by rock and soil deep within Earth
4. Which form of energy can travel through a vacuum?
   - (a) thermal
   - (b) solar
   - (c) mechanical
   - (d) chemical
5. What is the major process of surface rock formation on volcanoes?
   - (a) Rock cools quickly from melted rock.
   - (b) Rock is recrystallized by extreme pressure.
   - (c) Rock solidifies slowly deep underground.
   - (d) Rock is formed from deposited sediment.

True / False (10 marks)

Answer True or False

6. True or False: A toy truck will roll faster on a sandy surface than on a smooth surface.
7. True or False: Tidal effects of the Moon are the primary energy source for deep ocean currents that circulate large volumes of water.
8. True or False: A thermal insulator is a material that has a high specific heat capacity.
9. True or False: Matter has both mass and volume, which are essential properties of all substances.
10. True or False: A book moving horizontally on a flat table is likely due to an unbalanced force acting on it.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the role of chromosomes in human reproduction, specifically explaining what occurs when male and female gametes combine to form offspring.
12. Discuss how wind drag influences the characteristics of ocean surface currents, including their formation, movement, and impact on marine ecosystems.

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